The Fusion of Physical, Textural, and Morphological Information in SAR Imagery for Hurricane Wind Speed Retrieval Based on Deep Learning

Shanshan Mu, *Member, IEEE*, Xiaofeng Li, *Fellow, IEEE*, and Haoyu Wang

***Abstract*— This study developed a deep-learning-based model to retrieve sea surface hurricane winds from synthetic aperture radar (SAR) imagery. We introduce the essential idea, residual learning, of the Residual Net into the artificial neural network and design a deep cross-layer concatenation network (DCCN). The model inputs include SAR measured physical parameters in backscattering energy, the texture feature represented by the gray level co-occurrence matrix (GLCM), and the morphological hurricane feature. We collected 45 satellite SAR images from Sentinel-1 (S1) over hurricane conditions. These images were divided into 39 and six for model development and independent testing. A total of 16 127 wind samples acquired from 39 SAR images and simultaneously measured by the stepped frequency microwave radiometer (SFMR) were collected as model tuning datasets, among which 80% and 20% were used for training and validation. Our validation results show that the deep-learning-based model achieved a correlation coefficient (CORR) and root-mean-square error (RMSE) of 0.98 and 1.72 m/s for wind speeds up to 75 m/s. We further applied the model to six independent SAR images. The model significantly outperformed two existing geophysical algorithms and one backpropagation neural network (BPNN) algorithm with the RMSE is 2.61 m/s and a CORR of 0.95. Moreover, statistical analysis in different wind speed regimes indicates that our model shows a stable performance improvement than comparable algorithms. The RMSE decreases 10%–70%, especially the reduction of RMSE is more than 45% at high wind speed (>42 m/s). Furthermore, adding an independent rainfall estimate to the deep-learning model further enhanced the wind retrieval algorithm.**

***Index Terms*— Deep learning, hurricane wind, synthetic aperture radar (SAR).**

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Shanshan Mu and Haoyu Wang are with the Key Laboratory of Ocean Circulation and Waves, Institute of Oceanography, Chinese Academy of Sciences, Qingdao 266071, China, and also with the School of Oceanography, University of Chinese Academy of Sciences, Beijing 100049, China.

Xiaofeng Li is with the Key Laboratory of Ocean Circulation and Waves, Institute of Oceanography, and the Center for Ocean Mega-Science, Chinese Academy of Sciences, Qingdao 266071, China (e-mail: xiaofeng.li@ieee.org).

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I. Introduction

HURRICANES can cause significant societal impacts and severe economic losses, and hurricane forecasts are crucial for mitigating these impacts, especially the intensity forecast, which requires accurate surface wind speed measurements [1]. Space-borne synthetic aperture radar (SAR) images have been widely applied to monitor hurricane winds because of its uniquely high spatial resolution and its capability to image the 2-D ocean surface [2]–[8]. In addition, the retrieved SAR wind field can be used to exhibit a hurricane surface wind structure and dynamically forecast its evolution [9].

Methods for measuring sea surface winds with SAR have been continuously updated and developed recently [10]. Traditionally, sea surface winds have been retrieved from SAR based on the normalized radar backscatter cross section (NRCS). For example, co-polarized SAR data (VV/HH) has been used in geophysical model functions (GMFs) such as CMOD for C-band SAR to establish the relationship among surface winds, NRCS, and radar incident angles [11]. However, studies have shown that the co-polarized SAR signal becomes saturated under extreme ocean conditions, limiting their capability for wind retrieval at high wind speeds [12]. In contrast, for the cross-pol (VH/HV) signal, there is a linear relationship between the sea surface wind and the cross-polarized NRCS [13]–[15]. For instance, Hwang *et al.* [16] proposed two C-band VH GMFs (H14S and H14E) with wind speed and incidence angles by dividing the data into different incidence angle bins and wind speed groups. In addition, Zhang *et al.* [17] developed the C-3PO model for hurricane wind retrieval using VH-polarized NRCS and the incidence angle. Recently, new prospects that use combined co- and cross-polarized SAR measurements for ocean wind retrieval have also been developed [18], [19]. For example, Mouche *et al.* [18] proposed a combination approach that uses the newly designed VH-polarized GMF (MS1A) together with CMOD5.N, a co-polarized GMF, to provide wind speeds in both the inner and outer hurricane core [10], [20].

Besides measuring NRCS, SAR also has the imaging capability to capture ocean surface texture information modulated by various sea surface processes, including wind and ice [21]. Recently, Zhou *et al.* [22] examined wind-induced textures in SAR imagery and found a linear relationship between the sea

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surface wind speeds and the entropy using a parameter derived from the gray-level co-occurrence matrix (GLCM) [23].

This study developed a deep-learning algorithm to simultaneously utilize physical and textural information for wind retrieval. Deep learning, a powerful technology recently emerging in the machine-learning field, has demonstrated its significant superiority over traditional physical or statistical algorithms for extracting information from images in many industrial applications and can provide fast computational speed, especially in the face of high data volume. In addition, it has been demonstrated to be applicable for retrieving marine elements from remote sensing imagery, such as oceanic internal waves, eddies, sea ice, and sea surface winds [24], [25]. Xu *et al.* [26] developed a deep-learning-based model to retrieve hurricane wind from radiometer and scatterometer data. They confirmed the strong nonlinear fitting ability of a deep neural network. Wang *et al.* [27] designed a set of deep convolutional neural networks (CNNs) for estimating the intensity of tropical cyclones (TCs) from Geostationary Satellite Imagery.

To improve the retrieval accuracy of hurricane wind speed to the fullest extent, we deeply mined the adequate hurricane information in SAR images beyond the physical and textural information mentioned above. Furthermore, SAR images have also been widely used for hurricane structure or morphology studies, as the intense transient vorticity features and strong wind gradient can be observed near the inside edge of the hurricane eyewall from SAR images [4], [28]–[30]. Furthermore, TCs' central and outer sea areas have a low wind speed, while the TC eyewall usually has the strongest wind speed [4]. Therefore, this article also proposes to include morphological characteristics near the eye region for accurate wind retrieval.

Because of these strong correlations described above, three types of physical, textural, and morphological information from SAR hurricane images were fused into four deep cross-layer concatenation networks (DCCNs) to retrieve more accurate hurricane wind speed information. The remainder of this article is organized as follows: Section II describes the datasets used in this article; Section III presents the detailed introduction of the deep-learning model; the model results and discussion are shown in Section IV, including a performance comparison of models with different input parameters and also a comparison with existing physical models; and Section V summarizes the results.

II. Datasets and Data Matchup

For this study, 45 Sentinel-1 (S1) SAR images and the co-located airborne stepped frequency microwave radiometer (SFMR) measurements in a $\pm$2-h window were collected. In addition, three images of wind product collected by the Soil Moisture Active Passive (SMAP) radiometer were used for model validation.

A. SAR Data

The S1 images used in this study are C-band level-1 dual-polarization (VV/VH) ground range detected (GRD) SAR images covering 12 hurricanes [see Fig. 1] from 2016 to 2020 through the Satellite Hurricane Observation Campaign (SHOC) Project [18]. The dataset consists of 45 scenes: 30 Sentinel-1A (S1A) and 15 Sentinel-1B (S1B) acquired in interferometric-wide (IW) mode, and 13 in extra-wide (EW) mode. The swath width of IW mode and EW mode is 250 and 400 km, respectively, the incidence angles are 30°–46° and 18.9°–47.0°, and the pixel spacings are 10 m $\times$ 10 m and 40 m $\times$ 40 m. In addition, the IW data were resampled by spatial averaging to maintain the same pixel size as the EW data.

B. SFMR Data

We used the airborne SFMR hurricane wind measurements referenced at 10 m as ground truth to train and validate the DCCN model. SFMR data, provided by The National Oceanic and Atmospheric Administration (NOAA) Hurricane Research Division (HRD), includes wind speed and rain rate measurements along the aircraft flight track. Its spatial and temporal resolutions are 120 m and 1 s, respectively. The accuracy of SFMR data has been verified by the dropwindsonde measurements over a wide range of wind speeds (10–70 m/s), yielding a root-mean-square error (RMSE) below 4 m/s [31].

C. SMAP Data

The SMAP wind products were selected as the third-party data for triple-collocation analysis (TCA) with SAR retrieved wind speeds and SFMR data. The L-band ($\sim$1.4 GHz) SMAP radiometer, launched by the National Aeronautics and Space Administration (NASA) on January 31, 2015, is less affected by rain for TC winds observation than other instruments with higher microwave frequencies [32]. SMAP observations provide realistic wind retrievals, and have been widely used for TC measurements, since their wide swath ($\sim$1000 km). In this study, the SMAP winds were derived from the remote sensing systems (RSSs) [32], and they have a spatial resolution of 0.25°. And we found three SMAP wind images co-located with S1 and SFMR acquisitions within a $\pm$35-min time window.

D. Hurricane Track

This study uses two kinds of hurricane tracks for wind direction calculation and the geolocation adjusting of SFMR data, including the International Best Track Archive for Climate Stewardship (IBTrACS) dataset and the HRD-provided hurricane wind track (HWT) data.

IBTrACS, the most complete global best track dataset, was developed collaboratively with multiple agencies [33]. It is reported every 3 h and provides TC track (latitude and longitude), intensity (the maximum wind speed), and other parameters such as minimum Central Pressure. The IBTrACS dataset (the USA track) is mainly used for wind direction calculation through a revised inflow angle model [8], [34], and can also check the hurricane eye results for SAR imagery without covering the hurricane eye.

The HWT data, reported every 2 min, only provides the TCs' wind field center and time information and can be used with wind data from flights into the storm. Since most HWT

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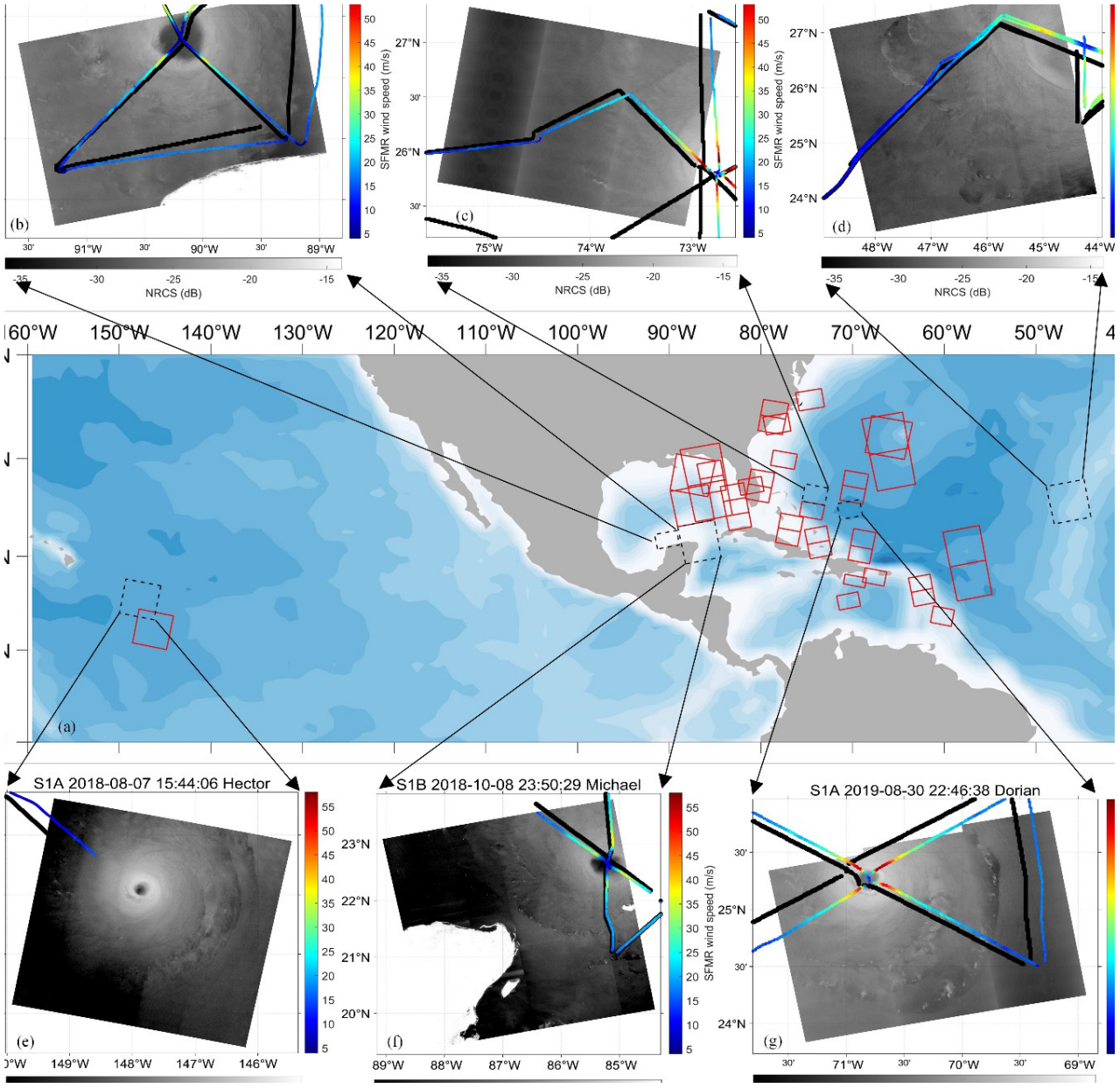

Fig. 1. Forty-five S1 images used for the DCCN model from 2016 to 2020. (a) Locations of the 45 S1 images. The map of collocated S1 NRCS in VH polarization acquired over Hurricane (b) and (f) Michael, (c) Florence, (d) Lorenzo, (e) Hector, and (g) Dorian, and tracks of SFMR observations before adjustment (black dots) and after adjustment (colored dots).

information is from aircraft observations, the same as SFMR wind data, and its high time resolution [35], we mainly rely on it to adjust the geolocation SFMR data. The HWT data, used in this article, are verified by SAR images covering hurricane eyes. In addition, we carefully checked the quality of the final matching data. Note that the IBTrACS data also provide TC track information for Hurricane Lorenzo when HWT is missing.

E. Matchup Data

SAR and SFMR data were matched within 2 h of the SAR image acquisition. We shifted the SFMR measurements based on hurricane movement provided by best track data during these 2 h. Six examples are shown in Fig. 1(b)–(g) with black/colored dots depicting the SFMR tracks before/after the adjustment. Those colored dots also represent SFMR wind speeds.

Among 45 SAR images, we used 39 and six for model development and independent testing. We first obtained 16 127 matchup data pairs from the 39 SAR images and the collocated SFMR observations and randomly divided them into 80% and 20% for the model training and validation. The training data were used for model learning to fit the neural network's parameters, such as weights. And the validation data are used to monitor the training process and continuously tune the parameters. The six independent testing SAR images [see Fig. 1(b)–(g)] are selected at different hurricane categories to demonstrate the wind speed retrieval precision under various

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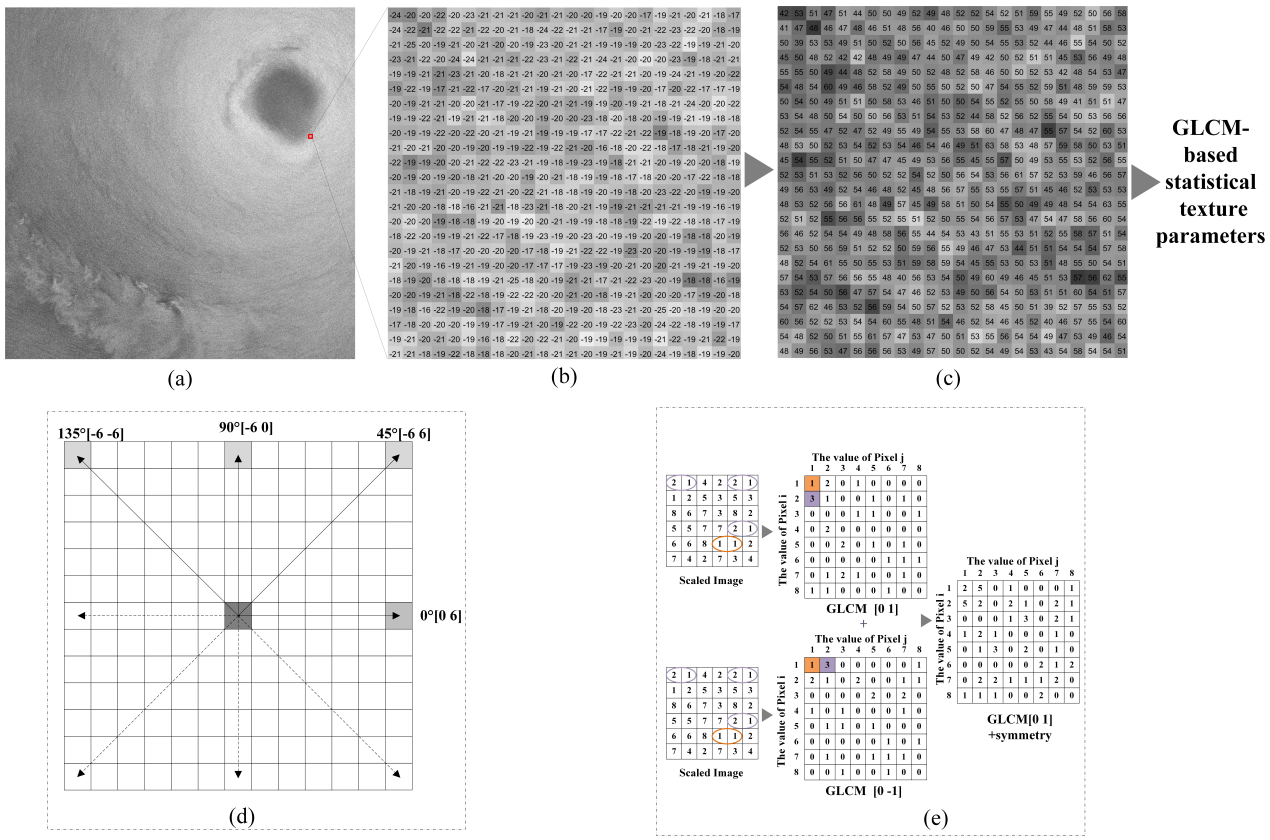

Fig. 2. Example of the flowchart calculation for GLCM-based statistical texture parameters. (a) SAR NRCS image in VH polarization, (b) subimage (24 × 24 pixels), (c) scaled image used for GLCM calculation, (d) distance and direction between the pixel of interest and its neighbor, and (e) example of the symmetrical GLCM calculation process.

sea conditions. The testing dataset consists of 3819 matchup data pairs to assess the generalization performance of the fully specified DCCN model compared with the existing two GMFs.

In addition, the SMAP measurements and SAR are collected within about ±35 min, and SMAP has a coarse resolution of about 25 km. Therefore, we ignored the hurricane motion for SMAP and SAR matchup, and did not adjust the geolocation of the SMAP data. We finally got three SMAP images acquired in August 31, 2019; September 29, 2019; and October 8, 2020, respectively.

III. Deep-Learning-Based Hurricane Wind Retrieval Model

In this section, we introduce the necessary process of model development including the extraction of input SAR parameters and the selection of input parameters and the setup of deep-learning-based framework.

A. Extraction of Input SAR Parameters

SAR provides three types of information: physical, texture, and morphological information that we introduced below.

1) Physical Parameters Retrieval: The SAR NRCS data processing includes radiometric calibration, thermal noise removal, and land masking using the European Space Agency (ESA) SNAP tool. To remove speckle noise, we performed spatial averaging of 24 × 24 pixels and reduced the spatial resolution to 960 m [7]. Then, we get the averaged SAR NRCS of VV and VH polarization (σ_0^{VV}, σ_0^{VH}) and the incident angle (θ) parameters.

2) Texture Parameters Retrieval: Apart from the physical information (SAR NRCS), SAR images also contain texture information reflecting spatial distribution characteristics. SAR texture involves the measurements of NRCS variation in pattern, shape, and size. In addition, it often indicates information concerning some physical processes and phenomena, such as TC and internal waves [22]. In our study, two statistical indicators of texture information, namely Entropy and Haralick's correlation, were implemented based on the most often used GLCM. Fig. 2 shows an example of the GLCM-based texture parameters calculation steps. And the detailed calculation steps are as follows.

To calculate the GLCM and GLCM-based SAR texture parameters, we first need to divide the original SAR image [see Fig. 2(a)] into a set of subimages [see Fig. 2(b)] with 24 × 24 pixels, the same subimages used for spatial averaging. Then, the subimages NRCS values (−40 to −15 dB for VH; −25 to 0 dB for VV) are rescaled to gray values between 1 and 64 since we set the GLCM parameter gray levels (N) at 64. Finally, the derived scaled images shown in Fig. 2(c) are used to calculate GLCM.

The GLCM indicates the frequency of co-occurring grayscale values in a given direction and at a certain distance in the scaled image [23]. In this study, we set distance as

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six pixels and orientation as 0°-45°-90°-135° to represent the horizontal adjacency between two pixels; it can be jointly expressed as [0 6; −6 6; −6 0; −6 −6] [see Fig. 2(d)]. For example, [0 6] means the number of rows and columns between the pixel-of-interest and its neighbor is 0 pixels and six pixels. Note that the number of rows is positive to the right and negative to the left of the pixel of interest, and the number of columns is positive downward and negative upward.

The GLCMs used to calculate texture parameters in our article are the Symmetrical GLCMs described by Haralick *et al.* [23]. Therefore, to be consistent with it, the last parameter to be set is the symmetry of GLCM about diagonal. That means, in the process of calculation, the two pairs of gray values (1, 2) and (2, 1) should be counted when calculating the number of times the value 1 is adjacent to the value 2. Therefore, for example, when the distance and direction are set to [0 6], the two GLCMs calculated in [0 6] and [0 −6] are added. To illustrate the calculation process more clearly, we show how the first three values in a GLCM are calculated in Fig. 2(e). For simplicity, we take a scaled image with gray values 1–8 as an example to calculate the symmetrical GLCM in the [0 1] direction.

After all these setups, each subimage will derive four GLCMs that indicate the relationship in four directions, and then each GLCM can be used to calculate GLCM-based texture parameters. Finally, we take the average of the texture parameters obtained by the four GLCMs as the texture parameters of the subimage.

GLCM-based statistical texture parameters, such as Entropy and Haralick's correlation, are essential to quantitatively analyze the texture of the images and describe the texture state [23]. The detailed calculation formulas are described below.

Entropy is a measure of the amount of image information, reflecting the complexity or nonuniformity of the texture in the image

$$\text{Entropy} = \sum_{i=1}^{N}\sum_{j=1}^{N} P(i,j)\log_{10}P(i,j). \tag{1}$$

Haralick's correlation indicates the similarity of GLCM elements in a row or column direction, reflecting the local gray-level correlation in the image

$$\text{Haralick}'\text{s correlation} = \frac{\sum_{i=1}^{N}\sum_{j=1}^{N} ijP(i,j) - \mu_1\mu_2}{\sigma_1\sigma_2} \tag{2}$$

where P(i, j) is the value of the pixel pairs (i, j) in the GLCM, and it specifies the number of times the pixel with a value i occurred horizontally adjacent to a pixel with value j. μ_1, μ_2, σ_1, and σ_2 are the means and standard deviations of the marginal distributions associated with P(i, j)

$$\mu_1 = \sum_{i=1}^{N}\sum_{j=1}^{N} i \times P(i,j) \tag{3}$$

$$\mu_2 = \sum_{i=1}^{N}\sum_{j=1}^{N} j \times P(i,j) \tag{4}$$

$$\sigma_1 = \sum_{i=1}^{N}\sum_{j=1}^{N} (i-\mu_1)^2 \times P(i,j) \tag{5}$$

$$\sigma_2 = \sum_{i=1}^{N}\sum_{j=1}^{N} (j-\mu_2)^2 \times P(i,j). \tag{6}$$

3) Morphological and Geographical Parameters Retrieval: The hurricane center locations for SAR images are derived using the template matching and particle swarm optimization algorithm [36]. This algorithm also obtains the center location of the hurricane in SAR images, only covering the rain band portion of the hurricane but not the entire eye. After determining the center position of all the hurricane images, we calculate the distance between each pixel point location and the hurricane center latitude/longitude location, expressed by R, in km. The Coriolis parameter $f = 2\,\Omega\sin\beta$. Ω is the angular velocity of the earth (7.292×10^{-5} s^{-1}), and β is the latitude.

B. Selection of Input Parameters

As we know, the selection of input parameters is crucial for the success of the wind speed retrieval model. After a series of data preprocessing mentioned above, as shown in the left panel of Fig. 3, we obtain five different types of parameters, including the rain rate and physical, textural, morphological, and geographical features.

After statistical correlation analysis of the matchup training data and based on previous studies, a total of 11 parameters are selected as the input of the model in this article. Four of them are physically meaningful SAR parameters based on the physical relationships summarized by previous studies, including σ_0^{VV}, σ_0^{VH}, θ, and wind direction. The physical information, σ_0^{VH}, has the highest correlation with wind speed. In addition, the Coriolis parameter f, which can affect the formation of TCs, was used as a geographically meaningful input parameter. Overall, the hurricane wind speed is a function of distance to the hurricane's eye. Therefore, the radius (R) was selected as the morphological input parameter. The four texture feature parameters are the GLCM-based Entropy and Correlation of SAR imagery with VV and VH polarizations which were proved to have a significant correlation with wind speed [22]. Previous studies have demonstrated that the highest wind regions within hurricanes are usually accompanied by significant rain [37], [38]. Therefore, rainfall information is also necessary for estimating its influence on radar backscatter [39]. Since the rain rate is also a parameter measured by SFMR, it was added to our model as an input parameter.

To evaluate the effects of different input parameters on the model, we designed four DCCN models with varying inputs according to the type of parameters. Among them, DCCN-1 has four most common physical parameters, σ_0^{VV}, σ_0^{VH}, θ, and wind direction, which were used in previous studies. DCCN-2 added four texture parameters based on DCCN-1, DCCN-3 added morphological parameters and the Coriolis parameters f based on DCCN-2, and DCCN-4 added SFMR rainfall data based on DCCN-3. We want to note that the DCCN-4 model was designed separately to evaluate the influence of

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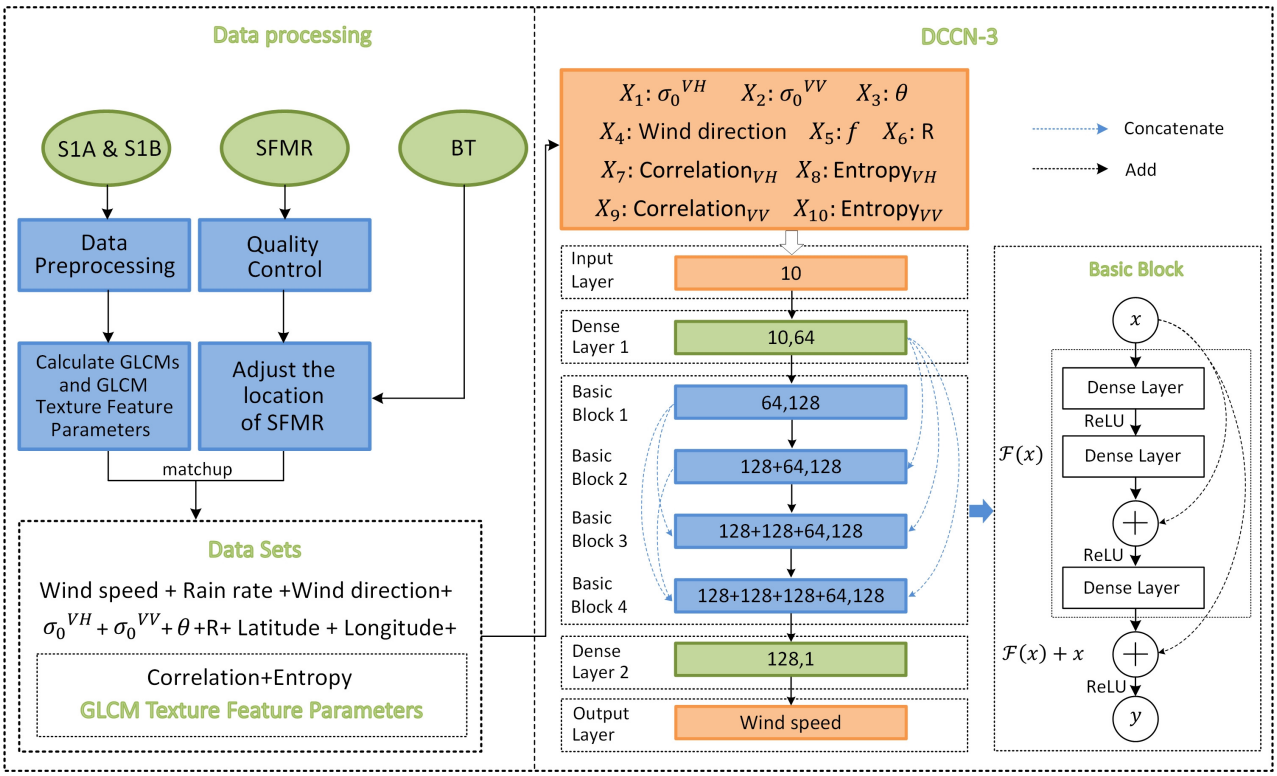

Fig. 3. Flowchart of the proposed method for deriving hurricane wind speed from S1 SAR data based on the DCCN (DCCN-3). S1A = Sentinel-1A; S1B = Sentinel-1B; SFMR = stepped frequency microwave radiometer; BT = best track; GLCM = gray level co-occurrence matrix; θ = incidence angle; σ_0^{VV} and σ_0^{VH} = normalized radar cross section with VV and VH polarization; R = the distance to the hurricane eye; f = the Coriolis parameter; x and y = the input and output of basic block; $F(x)$ = the residual mapping relation.

the rain rate information, which SAR does not observe, on the performance of the DCCN model.

C. Introduction of Model Structure

Given that 11 parameters are selected as the model inputs after the above process, deep-learning-based framework for the inversion of hurricane wind speed. Since it can flexibly fit all these different data types and obtain the optimal model by continuously optimizing the weight coefficients at a fast computational speed [25].

The network architecture of the deep-learning model is also a key factor affecting the model's performance. Therefore, we designed a deep-learning framework by adopting two essential ideas, one is "cross-layer concatenation," and the other is "cross-layer addition." The right panel of Fig. 3 shows the overall structure of the DCCN-3 model and the structure of the residual module. And the detailed flowchart of the multilayer neural network DCCN-3 with neurons and equations of the layer to layer transmission is shown in Fig. 4.

Fig. 3 shows the seven-layer structure of the DCCN model consisting of one input layer, two dense layers, and four basic blocks. A dense layer is the most frequently used deep-connected neural network layer. The numbers shown in the model structure indicate the number of neurons. For example, 10 indicates the number of neurons in the input layer. And the two numbers separated by commas in the middle-dense layers and basic blocks, such as 10 and 64, represent the number of input and output neurons. Adding numbers to the left of the comma, such as 128 + 64, means that the outputs of the above layers are input to this layer through "cross-layer concatenation," and the sum of the numbers is the input parameters layer.

The internal structure of DCCN-3 is described in detail through neurons in Fig. 4. Colored circles in different layers denote the neurons. The numbers to the left of the colored circle indicate the number of neurons, and X_i (i = 1, 2, 3, 4, 5, 6) and Y to the right of the colored circle represents the initial output of the current layer before concatenation. The bold plus sign in Layers 3–5 indicates that two or more output results will be passed into the next layer together. These formulas in the upper right corner of Fig. 4 show the calculation between layers. $F_{1,2}$ and $F_{6,7}$ denote the operation in the dense layer (Layers 2 and 7). $F_{2,3}$, $F_{3,4}$, $F_{4,5}$, and $F_{5,6}$ are the calculation process in the basic blocks (Layers 3 and 6). The parameters $\omega_{ij}, \omega_i, b_j$, and b in the formula represents the weight data and biased value used in deep-learning to optimize the model. Besides that, the basic block structure represents a residual mapping process. The four basic blocks go through the process shown in the dotted box in Figs. 3 and 4. X represents the input data from the upper layer in the basic block, and y is the output. And there are three dense layers in each basic block. We use $F(x)+x$ to represent the mapping relationship between input and output in the basic block, so $F(x)$ represents the residual mapping relation between input and output [40]. Finally, Relu represents the

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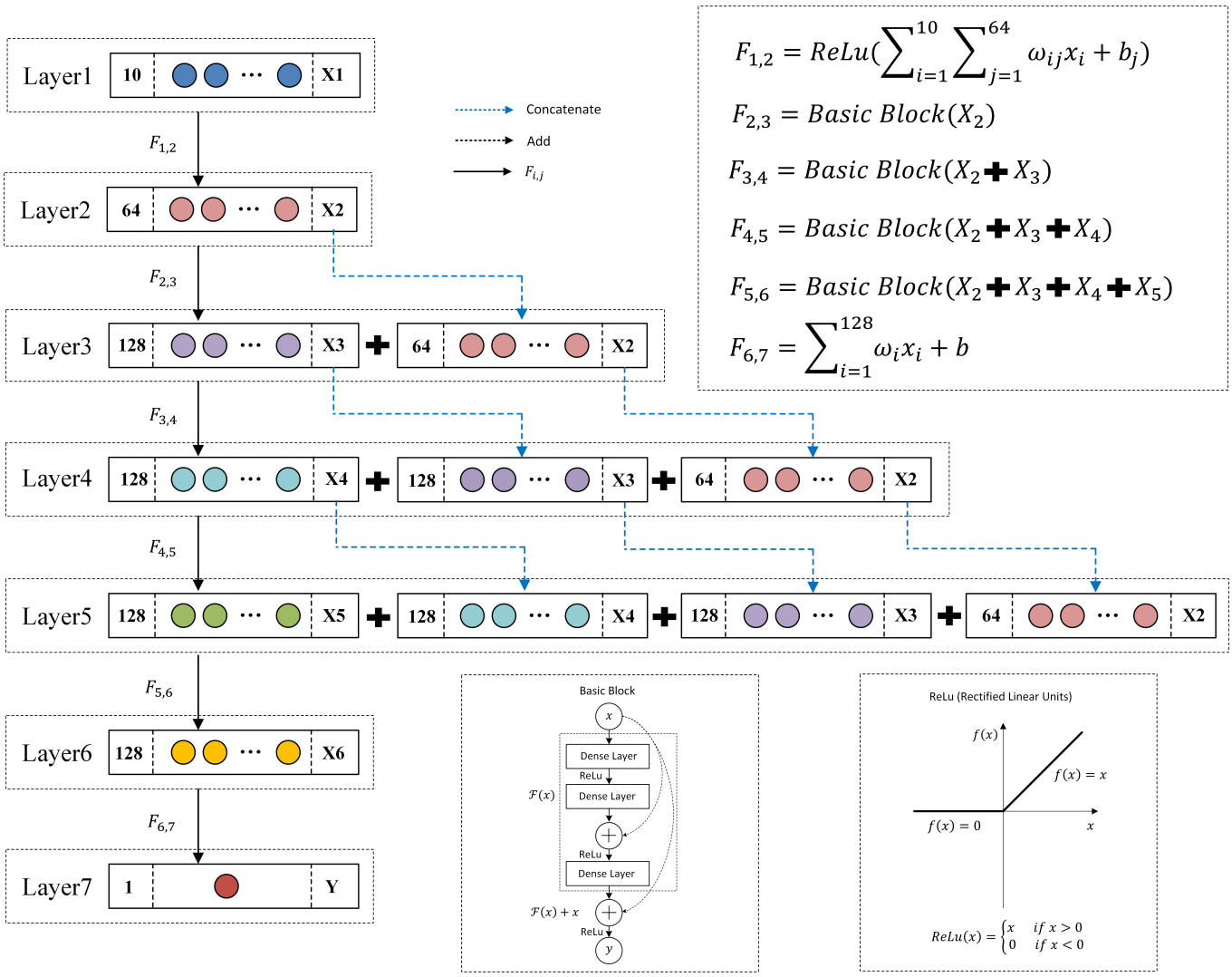

Fig. 4. Flowchart of the DCCNs (DCCN-3). X_1–X_7 = the outputs of the current layer before concatenation; Y = the output of Layer 7; $F_{1,2}$–$F_{6,7}$ = the operation formula between two layers; x and y = the input and output of basic block; $F(x)$ = the residual mapping relation.

activation function noted in the dotted box in the lower right corner.

The cross-layer concatenation method is to stack the outputs across one or more layers. This process does not involve an addition operation; that is, the original outputs of this layer will not be changed, but the data results of the two layers (the current layer and the previous cross layer) will be jointly input to the next layer. Therefore, as shown in Fig. 4, the number of neurons of the current layer will become the sum of the two layers. This design mainly avoids the loss of information in the feed-forward propagation process of the model. We know the input data are propagated from the first layer to the next layer. Each network layer can be approximated as a mathematical operation, including dozens to hundreds of weight coefficients. The results will also be processed by the activation function Relu, a piecewise linear function that changes all negative values to 0, and the positive values remain unchanged. Therefore, there will be a certain degree of information loss. The "cross-layer concatenation" process can precisely improve this problem by continuously splicing the output of each layer to each subsequent layer.

The second idea is cross-layer addition, which refers to "residual learning" in the classic CNN: the Residual Network (ResNet) [40]. The cross-layer addition process is located in submodules, called basic blocks. This idea can be realized by shortcut connections in feed-forward neural networks [40]. In short, the original outputs of the current layer will be added with the outputs of the previous cross layer, and then the added outputs will propagate to the next layer. It is noteworthy that this addition will not affect the number of neurons of the current layer and has been proven to be effective in solving the degradation problem caused by the increase of network depth. The addition also achieves a faster convergence speed and higher model accuracy, especially for deep networks [40].

In addition to the setting of the overall framework of the model, the mean square error (MSE) for the model outputs and the ground truth was selected as the loss function. The MSE is more suitable for the gradient-descending-based optimization algorithm used in this article. As the error decreases,

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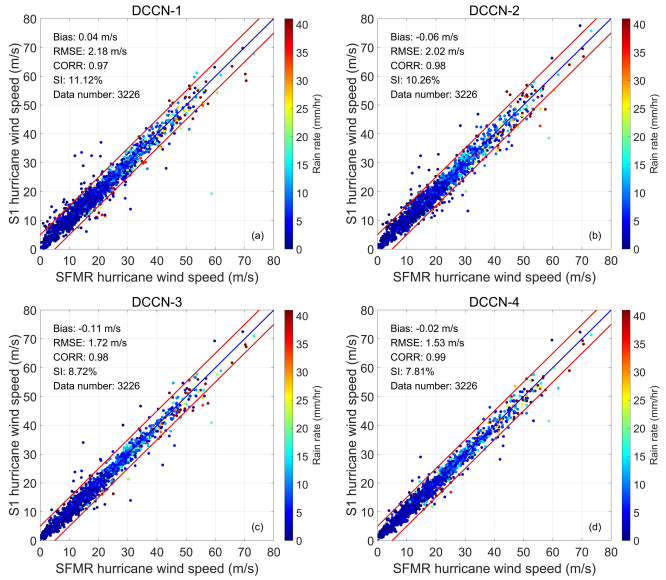

Fig. 5. Scatter plots of SAR-derived wind speeds through the DCCN-1 model (a) DCCN-2, (b) DCCN-3, (c) DCCN-4, and (d) SFMR observed wind speeds. S1 = Sentinel-1; SFMR = stepped frequency microwave radiometer; RMSE = root-mean-square error; CORR = correlation coefficient; SI = scattering index. The DCCN-1 model had four physical parameters, including σ_0^{VV}, σ_0^{VH}, wind direction and θ, DCCN-2 added four texture inputs, including Correlation_{VH}, Entropy_{VH}, Correlation_{VV} and Entropy_{VV}, DCCN-3 added morphological parameter R and Coriolis f, and DCCN-4 added SFMR measured rain rate.

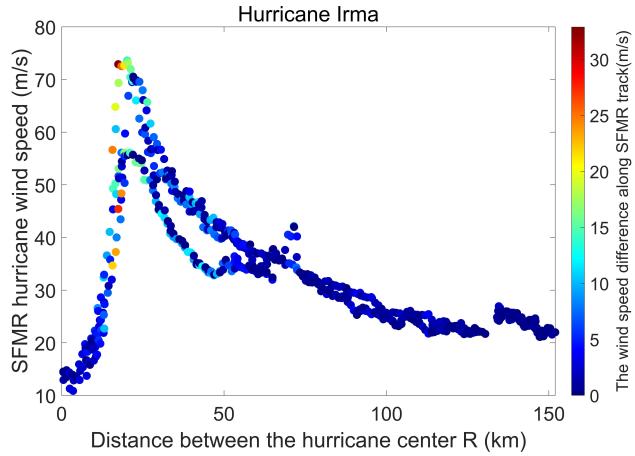

Fig. 6. SFMR hurricane wind speed as a function of distance from the hurricane center up to 200 km. The color bar indicates the wind speed difference from the previous adjacent point along the SFMR track.

the gradient also decreases, and the deep-learning model is conducive to convergence.

IV. Results and Discussion

A. DCCN Model Results With Different Inputs

This section compares the results of the four DCCN models with different input parameters proposed above on validation data. Fig. 5 shows a statistical comparison of the four DCCNs with SFMR hurricane winds up to 75 m/s. There is much more data with a wind speed range of 10–30 m/s than 40–80 m/s. Moreover, most data had a lower than 40 mm/h rain rate.

In general, all four algorithms provided reliable wind speed retrieval in the whole wind speed range up to 75 m/s, with most wind speed residuals within ±5 m/s bounded by the red lines. The experimental results show that using different inputs significantly impacts the performance of the wind speed estimate from the DCNN models, and adding more useful input parameters enhances the stability of the network performance.

The RMSE values for the four DCCN models are 2.18, 2.02, 1.72, and 1.53 m/s, respectively, and the correlation is 0.97, 0.98, 0.98, and 0.99. This improvement shows that DCCN models perform well and are robust. Fig. 5 shows that the distribution of scattered points is more concentrated around the blue solid line. The models' scattering index (SI) has decreased significantly, especially when adding morphological and texture parameters [see Fig. 5(c)]. The plots convincingly show the advantage of adding morphological features, geographical features, and textural feature parameters.

Fig. 5(d) shows that adding rain rate information further improves wind speed retrieval accuracy, with the decrease of RMSE by 11%. The improved performance of DCCN-4 with the addition of rainfall information indicates that the deep-learning model learns the influence of rainfall information on radar signals.

Taking DCCN-3 as an example, we find that more than 98% of data points had a bias (between DCCN-3 outputs and SFMR data) within ±5 m/s, and about 86% of the data had a bias within ±2 m/s. And 89% of the data points with an absolute bias greater than 5 m/s shown in Fig. 5(c) distribute within 50 km from the hurricane eye due to the residual geolocation matching error because the wind speeds around the hurricane eye have a large wind gradient causing the wind to change 60 m/s within 20 km [10]. Therefore, a small matching error may lead to significant NRCS and wind speed differences near the hurricane eye.

Taking Hurricane Irma (category-5 hurricane) as an example (see Fig. 6), a large wind speed gradient near the hurricane eye (within 25 km). Statistical analysis shows that the matching error of two pixels (1920 m) near the hurricane eyewall will cause an error of up to 8.5 m/s. We can conclude that a small matching error in the data will significantly impact the wind speed retrieval.

However, this retrieval error of wind speed caused by mismatch error cannot be entirely avoided because SAR data are generally considered to be obtained instantaneously. At the same time, SFMR observations usually last several hours, and the hurricane will continue moving during this period. Therefore, we adjust the position of SFMR data based on the interpolation of 6-hourly best track data. However, the adjustment process is uncertain because of the low temporal resolution of best track data and Hurricane movement and rotation uncertainty.

B. Independent Testing

For testing, we applied the DCCN models to the remaining six independent SAR images of hurricanes Lorenzo,

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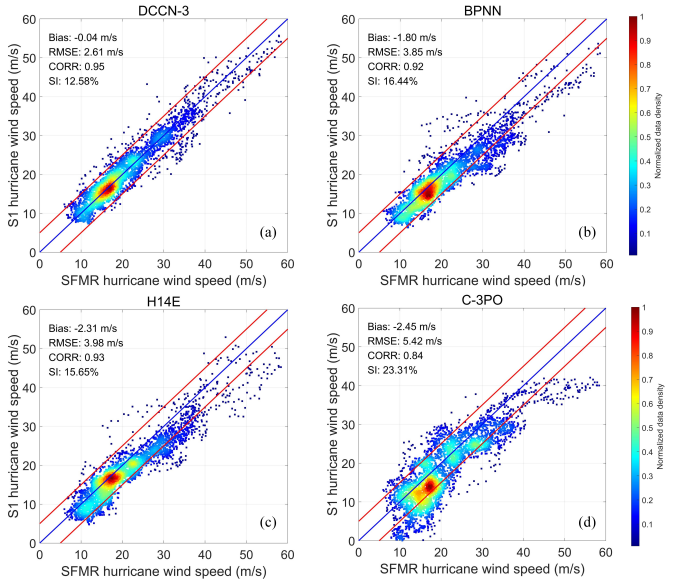

Fig. 7. Scatter plots of SAR-derived wind speeds from (a) DCCN-3 model, (b) BPNN model, (c) H14E [16], and (d) C-3PO [17] over Hurricane Lorenzo, Dorian, Delta, Hector, and Michael. S1 = Sentinel-1; SFMR = stepped frequency microwave radiometer; RMSE = root-mean-square error; CORR = correlation coefficient; SI = scattering index.

Dorian, Delta, Hector, and Michael [see Fig. 1(b)–(g)]. These SAR images represent different hurricanes intensities, including Category-1 [see Fig. 1(b)], Category-2 [see Fig. 1(d) and (f)], Category-3 [see Fig. 1(c)], and Category-4 [see Fig. 1(e) and (g)], according to Saffir-Simpson Hurricane Wind Scale. Since these data were not used in Section IV-A, we denoted independent testing data (ITD) sets for independent verification. Fig. 5 shows that the DCCN-4 has the best statistical parameters than the other three models because it adds the rainfall as an additional external parameter provided by SFMR. Therefore, we ultimately selected DCCN-3 for testing because it has the best statistical parameters among the first three models, and all these parameters are provided by SAR alone.

We performed two types of comparisons, one against physically-based GMFs H14E [16] and C-3PO [17] and another one against the well-known backpropagation neural network (BPNN) algorithm with the number of hidden layer neurons to 8, 12, and 5. The Hurricane wind speeds retrieved from ITD using the DCCN-3 model, BPNN model, and H14E against the collocated SFMR observations are shown in Fig. 7(a)–(d).

Overall, all four models provided reliable hurricane wind speed estimations below 40 m/s. In addition, DCCN-3, BPNN, and H14E models performed better with RMSEs less than 4.00 m/s. The C-3PO model does not work as well as the other three models, especially at high winds (>40 m/s) since it designed linearly between σ_0^{VH} and wind speed at a specific incidence angle and saturated at high winds up to 40 m/s [17].

DCCN-3, BPNN, H14E, and C-3PO had negative biases of 0.04, 1.80, 2.31, and 2.45 m/s, indicating that the four models underestimate the wind speed to a certain extent. And DCCN-3 had the lowest bias of −0.04 m/s. The four models have different RMSEs of 2.61 m/s (DCCN-3), 3.85 m/s (BPNN), 3.98 m/s (H14E), and 5.42 m/s (C-3PO). The results show a significantly better performance in DCCN-3.

Table I shows that the wind speed inversion results of the four models under different sea conditions are underestimated to a certain extent since most biases are negative. And the higher the wind speed, the greater the bias. Among them, DCCN-3 maintains the lowest biases among the four models under different wind speed categories, which are 0.39 m/s (U10 $\leq$ 17 m/s), −0.02 m/s (17 m/s < U10 $\leq$ 33 m/s), −1.67 m/s (33 m/s < U10 $\leq$ 42 m/s), −1.55 m/s (42 m/s < U10 $\leq$ 50 m/s), and −3.72 m/s (U10 > 50 m/s). Similar conclusions can be obtained by comparing the RMSE and CORR of the four wind speed retrieval models. DCCN-3 has the lowest RMSE and the highest CORR under the five categories of wind speed among all models. Moreover, statistical analysis in different wind speed regimes indicates that our model shows a stable performance improvement. The RMSE decreases 10%–70%, especially the reduction of RMSE is more than 45% at high wind speed (U10 > 42 m/s). Note the improvement of RMSE in the comparisons between DCCN-3 and the two published models, H14E and C-3PO. RMSE is reduced by at least 4 m/s in the high wind speed range (U10 > 42 m/s).

DCCN-3 obtains a CORR greater than 0.8 in the category of 17–33 m/s and >50 m/s. It shows that DCCN-3 performs better in these two wind speed regimes. And DCCN-3 also achieves a good CORR greater than 0.6 in the category of <17 m/s and 33–42 m/s. However, in the category of 42–50 m/s, DCCN-3 gets the lowest CORR, 0.25, possibly due to the significant fluctuation of CORR caused by the small amount of matchup data. Nevertheless, the RMSE is maintained well in this wind speed regime.

Fig. 8(a)–(f) shows that the DCCN-3 retrieved SAR wind maps overlapped with SFMR winds with the same color code to show the comparison more clearly. For comparison, we resampled the SFMR measurements matched with SAR IW and EW data by a spatial resampling at 3 and 5 km, respectively.

As shown in Fig. 8(a)–(f), DCCN-3 can obtain reasonable wind speed estimations under different sea conditions, reflecting the robustness of the designed deep-learning model. Fig. 8(e) and (f) has smaller hurricane eyes from the overall morphological characteristics and higher maximum wind speed than Fig. 8(a) and (c) [4]. Furthermore, by comparing the background color (SAR winds) and the color of the overlying scatters (SFMR winds), the agreement was even more evident outside the eyewall than that near the eyewall due to the most significant wind speed gradient around the hurricane eyewall [10].

C. Triple-Collocation Validation

In order to assess the errors of the SAR derived wind speed with the unknown "true wind fields" and further validate our model, we do a triple-collocation validation experiment to use the triple collocated dataset [41], [42]. The SMAP wind

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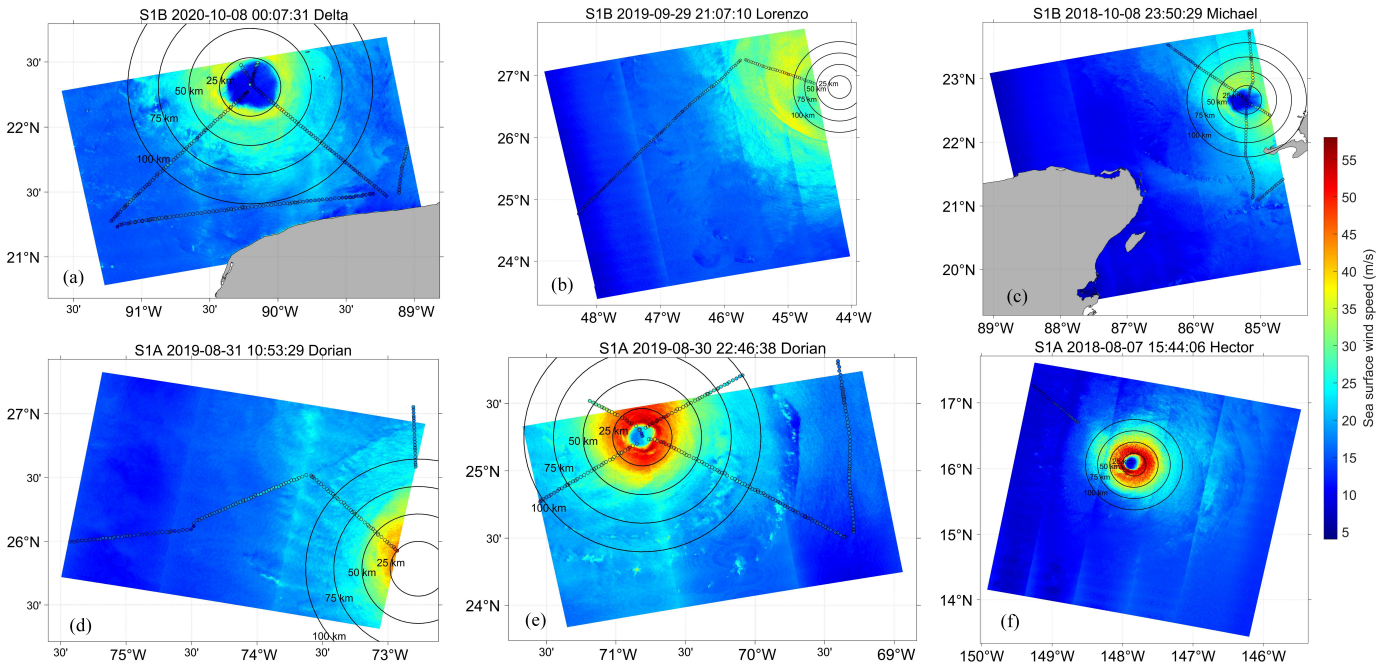

Fig. 8. DCCN-3 model retrieved sea surface wind fields over (a) Hurricane Delta, (b) Lorenzo, (c) Michael, (d) and (e) Dorian, and (f) Hector. The colored circles along the SFMR track indicate the collocated SFMR winds. The four concentric circles marked with 25, 50, 75, and 100 km represent the radius from the hurricane center.

TABLE I

PERFORMANCE OF THE FOUR WIND SPEED RETRIEVAL MODELS UNDER DIFFERENT WIND REGIMES

Category	Model	N	Bias (m/s)	RMSE (m/s)	CORR	SI (%)
U10 ≤ 17m/s	DCCN-3	1549	0.39	2.26	0.68	16.40
	BPNN	1549	-0.51	2.62	0.63	18.93
	H14E	1549	-1.15	3.12	0.59	21.39
	C-3PO	1549	-1.85	5.28	0.28	36.46
17 m/s < U10 ≤ 33 m/s	DCCN-3	1891	-0.02	2.50	0.87	10.91
	BPNN	1891	-2.08	3.89	0.74	14.39
	H14E	1891	-2.57	3.70	0.82	11.63
	C-3PO	1891	-2.09	4.88	0.70	19.29
33 m/s < U10 ≤ 42 m/s	DCCN-3	292	-1.67	4.12	0.62	10.42
	BPNN	292	-5.60	6.69	0.58	10.09
	H14E	292	-5.41	6.59	0.61	10.41
	C-3PO	292	-5.70	6.67	0.55	9.590
42 m/s < U10 ≤ 50 m/s	DCCN-3	50	-1.55	3.68	0.25	7.30
	BPNN	50	-4.44	6.06	0.15	9.00
	H14E	50	-5.21	7.89	-0.01	12.93
	C-3PO	50	-7.57	8.14	-0.01	6.50
U10 > 50m/s	DCCN-3	37	-3.72	4.13	0.80	3.33
	BPNN	37	-7.73	8.09	0.72	4.46
	H14E	37	-9.58	10.14	0.37	6.21
	C-3PO	37	-13.96	14.17	0.37	4.51

Note. RMSE=root-mean-square error; CORR=correlation coefficient; SI=scattering index.

products were selected as the third-party data for triple collocation with SAR retrieved wind speeds and the SFMR data. The validation results between the triple collocated dataset and the unknown "true values" including the calibration coefficients (α) and the error variance (EV) and the correlation coefficients (CC) are shown in Table II.

As can be seen from Table II, SMAP wind products are shown to have the largest EV, followed by the SFMR data. The wind speed derived from SAR data through our model achieves the minimum EV and the highest CC. The model developed in this article proved the most accurate through triple-collocation validation.

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TABLE II
TCA

Category	SFMR	SAR	SMAP
α	1.00	0.98	1.30
EV	2.43	1.96	2.85
CC	0.95	0.96	0.91

D. Discussion

This study contributes to the ocean surface wind retrieval under hurricane conditions by synergistically fusing eleven different parameters. The advantage of the deep-learning model over the traditional physical, statistical or empirical models is due to the complex structures and the unique property of neural networks to find the nonlinear relationships among parameters in an extensive database. The deep-learning model is like a black box that performs a highly complex mathematical fitting process. Although we cannot know the specific and detailed process in the black box, the more relevant input parameters we consider, the better the model's performance. In addition, the fitting process does not need artificial intervention.

The structure of the deep-learning model and the hyperparameters setting (layers, neurons, and optimization parameters) will lead to model performance differences. For example, our DCCN model [see Fig. 7(a)] is more complex than the classic BP network [see Fig. 7(b)]. The critical point in developing the DCCNs is to increase cross-layer information connection, effectively reducing the information loss in model forward propagation (FP). At the same time, the BP network is only a simple feed-forward neural network that cannot avoid the loss of information.

V. Conclusion

This article developed a deep-learning-based model, DCCN-3, for wind speed retrieval, especially for high-wind conditions. DCCN-3 fuses three types of physical, textural, and morphological information from SAR hurricane images and accurately estimate wind speed with an RMSE of 1.72 m/s (validation data) and 2.61 m/s (ITD) for wind up to 75 m/s. Compared with the existing physical models, the DCCN-3 model improved hurricane wind speed evaluation by adding morphological, geographical, and texture parameters, reducing RMSE by 1.37 m/s (compared with H14E) and 2.81 m/s (compared with C-3PO) over ITD. This study manifests the powerful data fusion ability of deep learning and fully confirms the potential of deep learning for wind speed retrieval under extreme weather conditions.

We also evaluate the improvement effect of rainfall on the model. Although the rain rate is from external SFMR data that only exists along aircraft tracks, it can be seen that the addition of rainfall data can further improve the hurricane wind speed retrieval to an RMSE of 1.53 m/s. However, precisely because SAR cannot provide the rain rate information at present, and our purpose is to derive the wind field from SAR directly, so we chose DCCN-3 instead of DCCN-4 as the final model of this article.

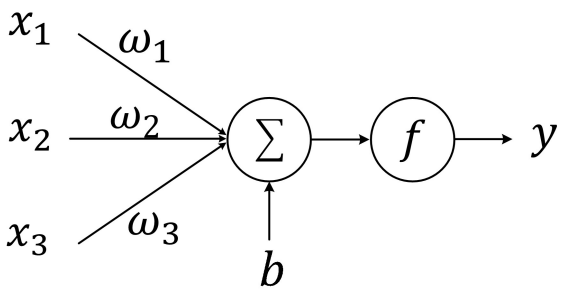

Fig. 9. FP in neural networks.

Appendix A
Information Loss in Feed-Forward Neural Networks

In the FP of the neural network, each neuron first makes a linear combination of the upper network neurons' outputs and then makes a nonlinear transformation through an activation function as the output of the current neuron. For example, the structure of neurons is shown in Fig. 9.

Outputs x_1, x_2, and x_3 generated by the upper neurons are transmitted to the current neuron as inputs. They are linearly combined under the action of weight terms (ω_1, ω_2 and ω_3) and a bias term b, and then through the nonlinear function f, and we get the result

$$y = f(\omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 + b). \tag{a1}$$

Then, we use the Relu activation function to activate the results of FP nonlinearly. The formula of the Relu function is

$$\text{Relu}(x) = \max(0, x). \tag{a2}$$

The above is the process of FP. We note that the negative part of y will be returned to 0 through the Relu function, which is bound to cause inevitable information loss. And the weights w_1, w_2, and w_3 will update through the model training to find the optimal weights for fitting the model. While these weights are difficult to fit the ideal relationships between the input and the output, they will also cause the loss of information in the process. To sum up, there will be a loss of information between layers in the process of FP.

Appendix B
Residual Learning

The concept of "residual learning," proposed by He *et al.* [40], is used to address the degradation problem in neural network training and learning. Usually, the network aims to find the potential mapping relationship $H(x)$ between input x and output y through training. However, we designedly add input x to the result before the last step (activation function, Relu) of the network through shortcut connection aiming to explicitly let above layers approximate a residual function, $F(x) = H(x) - x$, as the basic block section shown in Fig. 4. So, the learning goal of the module becomes to learn residual functions concerning the layer inputs instead of learning unreferenced functions $H(x)$. Simply put, residual learning can be understood as the process of fitting the residual $(H(x) - x)$, and this process

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has been proven to be effective in solving the degradation problem. We applied the idea "residual learning" twice in the basic block before and after the last dense layer.

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Shanshan Mu (Member, IEEE) received the B.S. degree in remote sensing science and technology from the Shandong University of Science and Technology, Qingdao, China, in 2015, and the M.S. degree in physical oceanography from the University of Chinese Academy of Sciences, Beijing, China, in 2020. She is currently pursuing the joint Ph.D. degree in physical oceanography with the Institute of Oceanography, Chinese Academy of Sciences, and the University of Chinese Academy.

Her research interests include synthetic aperture radar (SAR) applications in ocean surface wind and wave, artificial intelligent oceanography, and tropical cyclone.

Xiaofeng Li (Fellow, IEEE) received the B.S. degree in optical engineering from Zhejiang University, Hangzhou, China, in 1985, the M.S. degree in physical oceanography from the First Institute of Oceanography, Qingdao, China, in 1992, and the Ph.D. degree in physical oceanography from North Carolina State University, Raleigh, NC, USA, in 1997.

He was with the National Environmental Satellite, Data, and Information Service (NESDIS), National Oceanic and Atmospheric Administration (NOAA), College Park, MD, USA, from 1997 to 2019, where he was involved in developing many operational satellite ocean remote sensing products. He is currently with the Institute of Oceanology, Chinese Academy of Sciences, Qingdao. His research interests include synthetic aperture radar (SAR) applications in oceanography and marine meteorology, artificial intelligence oceanography, big data, and satellite image processing.

Dr. Li is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the *International Journal of Remote Sensing*; the Executive Editor-in-Chief of the *Journal of Remote Sensing*; and an Editorial Board Member of the *International Journal of Digital Earth*, *Big Earth Data*, and the *Journal of Oceanology and Limnology*.

Haoyu Wang received the B.S. degree in marine science and the M.S. degree in computer science and technology from the Qingdao University of Science and Technology, Qingdao, China, in 2019 and 2022, respectively. He is currently pursuing the joint Ph.D. degree in physical oceanography with the Institute of Oceanography, Chinese Academy of Sciences, and the University of Chinese Academy.

His research interests include El Niño-Southern Oscillation (ENSO) forecasting and estimation of oceanic subsurface variables using remote sensing and artificial intelligence techniques.

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